

ANOVA - 2

ANOVA with the R Language

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Figure 1: Arrange the arithmetic to compare more than two groups

ANalysis Of VAriance (ANOVA)

The analysis of variance is not a mathematical theorem, but rather a convenient method of arranging the arithmetic. R. A. Fisher (via Wishart 1934. Sppl. J. Roy. Soc. 1(1):26-61.)

1.1 Objectives

- The question of 1-way ANOVA
- Data and assumptions

- Graphing
- Test and alternatives
- Practice exercises

ANOVA - When?

anova-able scenarios:

- **one numeric continuous dependent variable** of interest
- **a factor that contains 2 or more levels**, often with a control
- When there are just two levels, it's essentially a t-test

ANOVA - scenarios

- **overall difference** At least one factor mean is different from the rest
- **Planned comparisons:** Targeted orthogonal contrasts
- **post-hoc tests:** compare any factors you want, with some correction.
- **Partition variation:** What's causing variance in the dependent variable?

Test statistic

- The test statistic for ANOVA is the **F ratio**
- It is the proportion of variance in the dependent variable between the groups, relative to that within the groups

Data and assumptions

- data example: +experiment in animal genetics, looking at the weight of chickens (8-week old weight in grams), where weight is the continuous dependent variable. The factor is the sire identity; the chicks were sired by one of 5 sires, thus sire is a factor with 5 levels A, B, C, D, and E.

Wide format data

```
A <- c(687, 691, 793, 675, 700, 753, 704, 717)
B <- c(618, 680, 592, 683, 631, 691, 694, 732)
C <- c(618, 687, 763, 747, 687, 737, 731, 603)
D <- c(600, 657, 669, 606, 718, 693, 669, 648)
E <- c(717, 658, 674, 611, 678, 788, 650, 690)
```

Long format data (preferred)

- Convert your data to long format (aov in R works with long data)

```
weight <- c(A,B,C,D,E)
sire <- c(rep("A", 8),
          rep("B", 8),
          rep("C", 8),
          rep("D", 8),
          rep("E", 8) )
new.long<-data.frame(weight, sire)
```

Assumptions of ANOVA

- Gaussian residuals (test graphically)
- Homoscedasticity (test graphically, both overall and within factors; residuals versus fitted plot)
- Independent observations (assumed)

Formal test of Gaussian residuals

- We can use the shapiro NHST `shapiro.test()`.

```
shapiro.test(rstandard(m1))
```

There is no evidence of difference to Gaussian in our residuals for our ANOVA model (Shapiro-Wilk: $W = 0.99$, $n = 40$, $P = 0.99$).

Homoscedasticity check

We will look at the residuals relative to the fitted values.

```
# Plot for homoscedasticity check
plot(formula = rstandard(m1) ~ fitted(m1),
      ylab = "m1: residuals",
      xlab = "m1: fitted values",
      main = "Spread similar across x?")
abline(h = 0,
       lty = 2, lwd = 2, col = "red")

# Make the mean residual y points (just to check)
y1 <- aggregate(rstandard(m1), by = list(new.long$Sire), FUN = mean)[,2]
# Make the x unique fitted values (just to check)
x1 <- unique(round(fitted(m1), 6))

points(x = x1, y = y1,
       pch = 16, cex = 1.2, col = "blue")
```

Bartlett test for equal variances between groups

- `bartlett.test()`: do groups have signif different variances?
- significant differences = homoskedasticity violated

```
bartlett.test(formula = weight~sire, data = new.long)
```

We find no evidence that variance in offspring weight differs between sires (Bartlett test: K-squared = 1.69, df = 4, P = 0.79).

Graphing ANOVA

Recommended: Boxplot, with some way to show the central tendency of the data separately for each factor level. For continuous variables, boxplots show this perfectly.

- For count variables, barplots are sometimes used with the height set to the mean, along with some form of error bar.

Graphing ANOVA

```
boxplot(Weight ~ Sire, data = new.long,
        ylab = "Weight (g)",
        xlab = "Sire",
        main = "Effect of Sire on 8-wk weight",
        cex = 0)

abline(h = mean(new.long$Weight),
       lty = 2, lwd = 2, col = "red") # Mere vanity

points(x = jitter(rep(1:5, each = 8), amount = .1),
       y = new.long$Weight,
       pch = 16, cex = .8, col = "blue") # Mere vanity
```

ANOVA basic output

- The ANOVA Table”
 - There are 2 rows; for the sire effect and the residual
 - The test statistic is the F value (1.87)
 - The P-value column is named “Pr(>F)” (0.14)
 - There are 4 degrees of freedom for the factor
 - Residual degrees of freedom is 35
- No sig evidence that sire explains variation in weight

Contrasts and *post hoc* test

- Let’s say that sire C is our reference level sire
- For now, we’ll use `lm()` to compare everything to sire C
- you can generate contrasts and make tests using `emmeans` e.t.c.
- Let’s interpret the coefficients
- What can we notice about the F value and p-value?

Post hoc tests

- The phrase *post hoc* implies an afterthought. Tread carefully
- When multiple tests are made on the same data, it increases the chance of discovering a false positive (Type I error)

- p-values can be adjusted to keep Type I error chance at 5%.

Bonferroni

- The **Bonferroni adjustment** of p-values:
 - Simply divides the P by the number of *post hoc* pairwise comparisons, so you end up evaluating at a lower P-value
 - It is considered conservative, and there are alternatives.
 - alternatives exists e.g. Tukey's e.t.c.
 - With `pairwise.t.test()`, the output will be a matrix of p-values

Tukey HSD (Honestly Significant Differences)

- The Tukey HSD test is less conservative Bonferroni
- We use the `Tukey.HSD()` function to apply it here.
- Also note that the p-values tend to be smaller for the exact same comparisons, but there are still no significant comparisons.

Alternatives amidst assumption violations

In case the assumptions of 1-way ANOVA cannot be met by the data, there are a few options:

- Transformations (`log()`, `sqrt()` e.t.c.)
- GLMs
- Non-parametric tests: a great escape, but tend to have less power

Kruskal-Wallis alternative to the 1-way ANOVA

```
# Try this:
# ?kruskal.test
kruskal.test(formula = Weight ~ Sire,
              data = new.long)
```

- The result is qualitatively the same as that for our 1-way ANOVA test

ANOVA calculation details

```
# Our data  
chicken.wide
```

ANOVA Equations

Equations

$$(1) \text{ Grand Mean} = GM = \sum_{i=1}^k \sum_{j=1}^n Y_{ij} \times \frac{1}{n_{total}}$$

$$(2) \text{ Sum of Squares Between} = SS_{between} = \sum_{i=1}^k n_i (\bar{Y}_i - GM)^2$$

$$(3) \text{ Sum of Squares Within} = SS_{within} = \sum_{i=1}^k \sum_{j=1}^n (Y_{ij} - \bar{Y}_i)^2$$

$$(4) \text{ Sum of Squares Total} = SS_{between} + SS_{within}$$

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ANOVA Variables

Variables

k = number of factor levels

n_i = number of samples for group i

n_{total} = total number of samples for all groups

Y_{ij} = observation j in group i

$\bar{Y}_i$ = mean of group i

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ANOVA Sources of variation table

Sources of Variation

<i>Source of Variation</i>	$ df$	$ Sum Sq$	$ Mean Sq$	$ F$	$ $
<i>Between Groups</i>	$ k - 1$	$ Eq (2)$	$ \frac{Eq (2)}{k-1}$	$ \frac{MeanSq_{between}}{MeanSq_{within}}$	$ $
<i>Within Groups</i>	$ n_{total} - k$	$ Eq (3)$	$ \frac{Eq (3)}{n_{total}-k}$	$ $	
<i>Total</i>	$ n_{total} - 1$	$ Eq (4)$			

Do an ANOVA “by hand” programmatically

- For the ANOVA code in the script, try to follow what is going on in the code
- It is okay if not every detail is clear yet
- Do we get the same answer as `aov()`?